Title: Using the power law size distribution to extrapolate and compare microplastic number and mass concentrations in environmental media

Key words: Contaminant, pollution, alignment, harmonization, atmosphere, ocean, soil, sediment, biota, seawater, freshwater, risk, nanoplastic, macroplastic

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ABSTRACT

Microplastic (MP) number concentrations in the environment increase exponentially with decreasing particle size. Studies reporting environmental MP concentrations typically do so for variable MP size ranges, depending on sampling, processing and analytical detection methods. This leads to difficulties in intercomparison and extrapolation of studies, which is critical for data reviews, plastic dispersion modeling, and environmental and human health risk assessment. In this study, we summarize the current understanding of the MP particle size distribution (PSD), based on the power law model. We highlight how standard linear regression of the power law slope is strongly biased by data binning, and show that fitting a cumulative PSD (C-PSD) removes the binning bias. The existing MP size-alignment framework is extended to C-PSDs to extrapolate observed MP number and mass concentrations to the full MP size range (1 to 5000 µm, noted $MP_{1\text{-}5000\mu m}$), or any other sub-size range. We confront the C-PSD power law model with 62 published ocean and atmosphere PSDs from the literature, compiled in the MPsizeBase open access database. Fitted power law slopes for fragments (-2.76±0.62) are steeper than for fibers (-1.84±0.38), reflecting fragmentation dimensionality. Among MP fragments, PSD slopes do not vary significantly between the atmosphere, surface and subsurface ocean. We further demonstrate that the large discrepancy between surface ocean MP concentrations measured by net tows and discrete, pumped samples arise primarily from their different minimum detectable sizes. After aligning datasets to a common size range, fragment concentrations converge to within a factor of 10. In contrast, fiber concentrations from net tows remain lower than those from pumped samples, consistent with sampling losses and detection limitations for elongated particles. Across all 62 MP PSD datasets analysed, size-aligned $MP_{1-5000\mu m}$number and mass concentrations are respectively 600x and 3x higher than reported concentrations, reflecting the high abundance of small particles predicted by the power law PSD. Together, these findings highlight that size extrapolation to a common range is essential to intercompare datasets and to distinguish environmental patterns from methodological artifacts.

INTRODUCTION

Microplastics (MPs) are typically defined as plastic particles with length ranging from 1 to 5,000 µm (Thompson et al., 2024). However, observations of MPs in the environment rarely cover this formal size range. More specifically, each combination of sampling and analytical detection methods permits quantification of a certain MP size range. In the surface ocean, a typical method for sampling MP is by plankton, neuston or manta net tows with a mesh size of around 300–330 μm retaining large MP above this cutoff. In air, high volume samplers can filter the thousands of m^3 needed to retrieve larger airborne MP. Initially, large MP identification in net tows or on high volume filter samples was done by manual sorting and counting under optical microscopes (Hidalgo-Ruz et al., 2012). The subsequent use of analytical tools such as spectroscopy and mass spectrometry has enabled the quantification of ever smaller MP particles sampled with smaller net mesh down to 25 μm or pumped filtration down to 10, 5, or even <1 μm. FTIR protocols have been developed to determine the number of MPs and identify polymers down to 10–20 μm (Nava et al., 2021; Rathore et al., 2023). Raman spectroscopy techniques lowered the limit of MP detection down to 1–2 μm (Nava et al., 2021). Finally, pyrolysis-gas chromatography-MS can identify and quantify the mass of all MP sizes, including nanoplastics (NPs) smaller than 1 μm; however, it does not provide detailed information on MP or NP particle size distribution. Since smaller MP (< 25 μm) are far more abundant than larger MP, smaller sample amounts are generally needed to observe them, permitting the use of in situ pumps of Niskin bottles to explore the deep ocean, and low volume airborne pumped samples to achieve daily time resolution. The observed MP size range for each sampling and detection method combination is dictated by the limit of detection (LOD) of the specific methods on the lower end, or by the sample volume and occurrence of less abundant, larger MPs on the higher end.

The variability in reported MP size ranges, alongside the exponential increase in MP count with decreasing MP size, complicates MP concentration intercomparison. Studies reporting an abundance of small MPs cannot be directly compared to those reporting a smaller number of larger MPs, as the difference in MP concentration reflects the underlying exponential particle size distribution (PSD) rather than the actual level of MP pollution. To make robust comparisons of MP pollution in the environment, MP data must be expressed on the same number or mass concentration scale, i.e. over the same particle size range. This is crucial for any concentration comparison, whether for estimating the mass balance of MPs in the environment, conducting risk assessment analyses, or determining environmentally relevant concentrations for exposure experiments.

Microscopy-based studies have repeatedly shown the exponentially increasing number of MP particles with decreasing size. The power law is the most commonly used model to describe the PSD of environmental MP, as first reported by Cózar et al. (2014) for global surface ocean MP observations. Since then, this model has been used to describe PSDs of MPs in water, sediment, soil, the atmosphere, and biota (Cózar et al., 2014; Kaandorp et al., 2021, 2023; Kooi et al., 2021; Kooi and Koelmans, 2019; Yakovenko et al., 2025). A formal PSD analysis framework based on the power law has been proposed and used to intercompare PSDs and extrapolate MP concentration data to default MP size ranges (Kooi et al., 2021; Kooi and Koelmans, 2019). This framework has been employed in human risk assessments (Koelmans et al., 2020; Yakovenko et al., 2025) and marine MP mass balance studies (Kaandorp et al., 2021). However, complications arise for PSDs that aggregate MP counts in binned histograms. Binned MP size histograms, reported by the majority of studies, lead to biased power law fits that depend on bin size (Leusch et al., 2023). As most studies do not provide their raw data, i.e. the size of each observed MP particle, wider use of the power law model to extrapolate and size-align MP concentrations is not possible.

Building on previous research (Kooi et al., 2021; Kooi and Koelmans, 2019; Leusch et al., 2023), in this study we examine bias in power law fitting due to variations in reported PSD bin size and propose and validate a new approach to fitting cumulative PSD data that removes binning bias. We propose a framework for extrapolating observed microplastic (MP) number and mass concentrations over a restricted range to any MP size range of choice (e.g. the usual 1–5000 μm MP size range). To this end, we utilise the recently compiled MPsizeBase database (Sonke et al., 2025b), comprising 40 microscopy studies reporting 62 MP PSDs, and 113 MP concentration data points for fibres and fragments in various environmental compartments, including the atmosphere (both indoor and outdoor), the surface ocean, and the deep ocean. We confront these PSDs with the power law model to gain insight into methodological artifacts, plastic fragmentation and particle size sorting processes. Finally, we examine PSD and size-aligned MP concentration variability in the natural environment.

The remainder of this paper is structured as follows. We first introduce the theoretical basis of power law PSDs and their relevance for microplastic fragmentation (Sect. 1). We then describe how microplastic concentrations and PSDs from different studies can be intercompared (Sect. 2), before examining how binning choices introduce systematic bias in power law fitting (Sect. 3). Using reference datasets, we introduce and validate an alternative approach to PSD fitting that removes binning bias. We subsequently address the conversion of MP number concentrations to mass concentrations (Sect. 4). Building on this methodological framework, we analyze environmental MP PSDs, focusing on the influence of detection limits, sampling strategies, and analytical methods on PSD properties (Sect. 5). Finally, we apply our approach to extrapolate and intercompare environmental MP concentrations across common size ranges (Sect. 6) and discuss extrapolation strategies that do not rely on cumulative PSD fitting (Sect. 7).

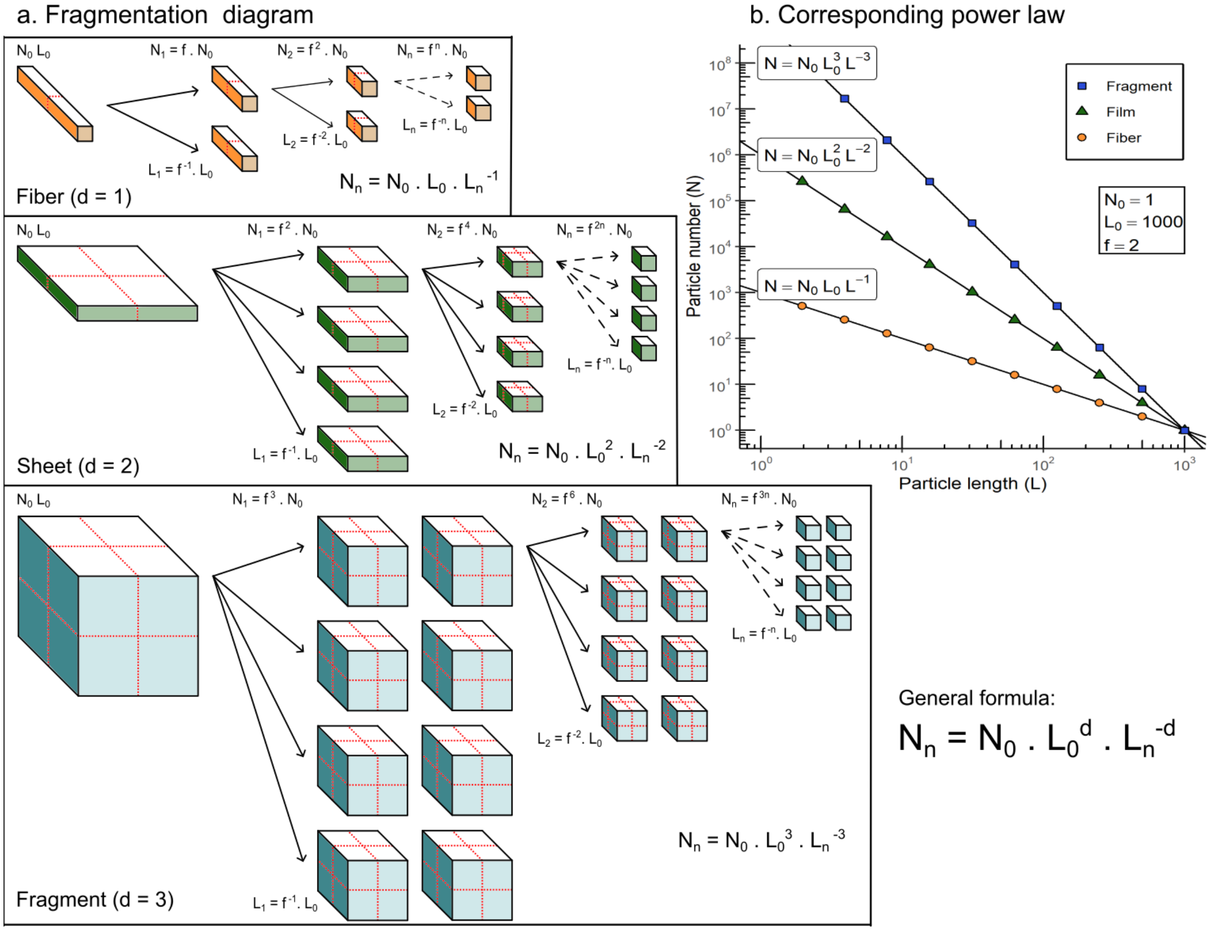

Figure 1*: Fractal plastic fragmentation model that produces ever smaller plastic and eventually microplastic (MP) particles, assuming that plastic breaks into smaller pieces in a self-similar manner, i.e., the size distribution follows a power law. a) Three MP shapes of simplified dimension, d, are considered: fibers (d=1), sheets, representing bottles, containers, bags, mulch and wrappings (d=2), and fragments, representing various 3-dimensional debris (d=3). N_n is the number of particles of size L_n. N_0 and L_0 is the initial number and size of the particles, respectively. b) Particle size distributions for fiber, film and fragment shapes. Image and model inspired by Cózar et al., 2014; Kaandorp et al., 2021; Turcotte, 1986.*

1. Power law theory

Natural and engineered materials fragment due to weakening of molecular surface structure by UV radiation, and chemical or enzymatic oxidation, followed by erosion along microcracks which is accelerated by physical abrasion and temperature oscillations. Such fragmentation of large plastic objects and debris produces smaller plastic particles, including MP and nanoplastic. The particle size distribution (PSD) of MP particles often follows a particular statistical relationship (the probability density function) between particle size, L (length), and abundance, N (number), called a power law (Kaandorp et al., 2021; Kooi and Koelmans, 2019; Leusch et al., 2023). The power law probability density function is:

$$N(L) = b.L^{a} \qquad \text{Eq.(1)}$$

where a and b are the slope and the intercept of the power law model, respectively. In a log-log space, Eq.(1) describes a straight line of slope a and intercept $log(b)$.

To understand why the power law appears during fragmentation processes, we build a simplified MP fractal fragmentation model based on Kaandorp et al., 2021 (Kaandorp et al., 2021). Figure 1.a illustrates the repeated fragmentation of three simplified plastic shapes of dimension d: fibers ($d=1$), sheets, representing bottles, containers, bags, mulch and wrappings ($d=2$), and fragments, representing various 3-dimensional debris ($d=3$). At each fragmentation step, the main length of the particles is divided by $f=2$, and the number of plastic particles increases exponentially in order to keep the total volume (and mass) of plastic constant. Because the volume expressions for fibers, sheets and fragments are proportional to L, L^2 and L^3 respectively, the number of particles during fragmentation increases faster for fragments, than for sheets, than for fibers with decreasing particle size. This can be visualized by plotting log(N) as a function of log(L) (Figure 1.b), which linearizes the power law so that the exponent a can be estimated from the linearized slope. Under these idealized fragmentation circumstances, the slope coefficient for the fiber (a = -1), sheet (a = -2), and fragment (a = -3) PSD corresponds to the dimension of the shape, d. Note that the power law distribution holds for any fragmentation factor f >1.

Table 1. *Fictional MP fragmentation example. Four samples are considered (Frag 1, Frag 2, Film, and Fiber), corresponding to the 3 MP shapes considered in this study (fiber dimension, d=1, film d=2 and fragment d=3). A particle size distribution (PSD) for each sample was generated following the power law distribution in Eq.(1) with $b=10^6$ and a=-d. Each PSD is deliberately sampled in a different size range, and the observed number concentration (MP# m^{-3}) is reported. Using the power law coefficients in Eq.(2), the extrapolated $MP^{\#}_{1-5000\mu m}$ number concentration is calculated. As samples Frag 1 and Frag 2 were generated with the same PSD, they have the same $MP^{\#}_{1-5000\mu m}$ concentrations. However, as their observation ranges were different, their original reported MP concentrations were different.*

Sample	Observation size range [µm]	Reported concentration [MP# m^{-3}]	Power law equation $b\,x^{a}$	Extrapolated $MP^{\#}_{1-5000\mu m}$ concentration [MP# m^{-3}]
Frag 1	1 - 10	4 950	1000 L^{-3}	5 000
Frag 2	50 - 250	2	1000 L^{-3}	5 000
Film	10 - 100	900	1000 L^{-2}	9 998
Fiber	250 - 1 000	29957	1000 L^{-1}	85 172

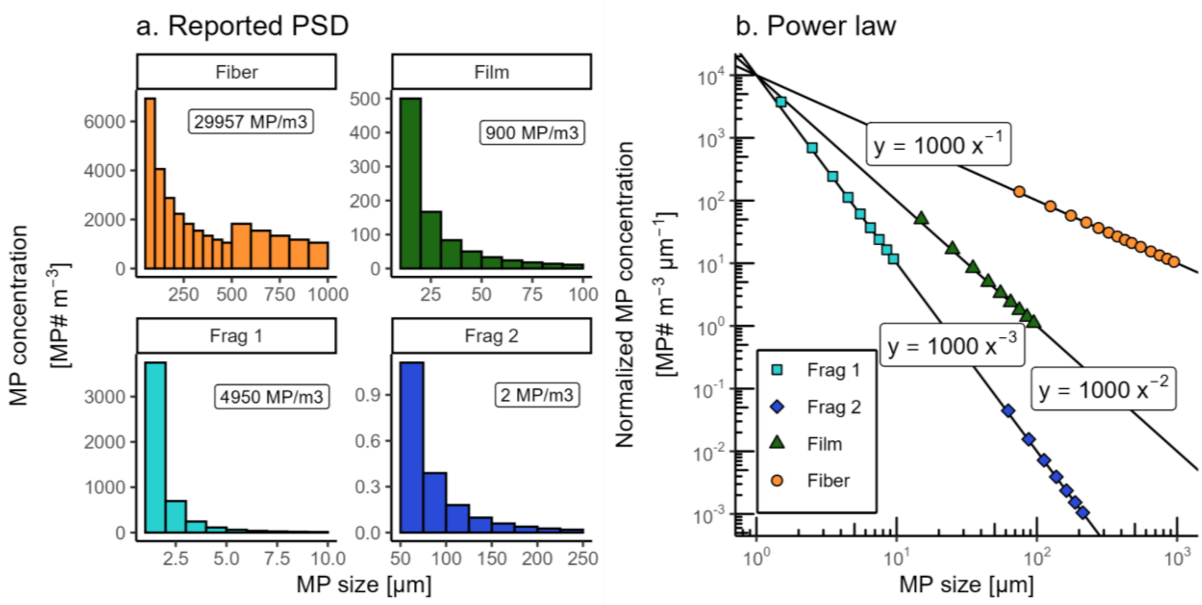

Figure 2 *: Fictional example of reported MP PSD in the literature (a.) and the corresponding power laws in bin-normalized MP number concentrations (b., MP# m^{-3} μm^{-1}). Fragment size range (Frag 1, Frag 2) is varied on purpose, and so is bin size for fibers (a).*

2. Intercomparing MP concentrations and PSDs

As indicated in the introduction, measurements of MP in the environment are often made over different MP size ranges, depending on the sampling and analytical techniques. In addition, studies tend to bin MP number observations in widely variable bin ranges to report PSD histograms. This variability leads to difficulties for intercomparison of studies, which is critical in MP science. Previous studies have suggested that binned PSD data can be normalized for bin size and extrapolated using the power law, so that different studies can be compared on an identical numeric basis (Kaandorp et al., 2021; Leusch et al., 2023). To illustrate this, we generate a fictional dataset of 4 samples whose distribution follows Eq.(1): two fragment samples of different measured size range, (a = -3), one for films (a = -2) and one for fibers (a = -1). The intercept is the same for all samples (b= 10^6). Figure 2.a shows how fragmentation produces different PSD histograms, and how the use of variable size bins (1, 10, 25, 50 µm) affects visualization of the PSD, with a wider bin size for fibers and films leading to an apparent increase in MP number concentration (MP# m^{-3}). In addition, the 4 chosen samples deliberately cover different MP size ranges, reflecting potentially different analytical techniques. This disparity implies that the MP concentrations of 29957, 900, 4950 and 2 MP# m^{-3} reported alongside the histograms are not directly comparable. Sample Frag 1 seems to be 2000 times more concentrated than sample Frag 2. To allow intercomparison of the measurements, we first normalize the reported MP concentration in each bin by dividing it by its bin size, and plot the log of the normalized MP concentration (MP# m^{-3} μm^{-1}) as a function of the log of MP size (µm, geometric mean of the bin boundaries, following Leusch et al., 2023 method (Leusch et al., 2023)) (Figure 2.b). The PSD thus obtained will be referred to as Bin-Normalized PSD (BN-PSD). Similar to Figure 1.b, we observe in Figure 2.b that MP shape affects the power law slope. For all shapes, fragmentation leads to more abundant small MP, but the effect is stronger for fragments, than for sheets, than for films, as seen in the more negative slope a.

In order to intercompare the MP number concentrations of the four normalized sample PSDs, we can use the power law to extrapolate the sample observations (data points in Figure 2.b) to a shared MP size range, such as the full MP size range (1-5000 µm) or any sub-range. To do so, we integrate the power law PSD, Eq.(1), between the chosen particle size limits:

$$MP^{\#}_{L_1-L_2\mu m} = \int_{L_1}^{L_2} N(L)\,dL = \int_{L_1}^{L_2} b.L^a\,dL = \frac{b}{a+1}.\left({L_2}^{a+1} - {L_1}^{a+1}\right) \quad \text{Eq.(2)}$$

Where $MP^{\#}_{L_1-L_2\mu m}$ [MP# m^{-3}] is the MP number concentration within size range L_1 to L_2 [µm]. b [MP m^{-3} μm^{-a}] and a [-] are the intercept and the slope of the power law model, as illustrated in Figure 2.b, respectively. L is the MP length [µm]. Note that in practice, the slope coefficient a should be lower than -1 for this integral to converge. Applying Eq.(2) to our 4 fictional examples for the size range of $L_1 = 1$ µm to $L_2 = 5000$ µm, we obtain the following directly comparable $MP^{\#}_{1-5000\mu m}$ concentrations of: 85 172 MP# m^{-3} for fibers, 9 998 MP# m^{-3} for films, 5 000 MP# m^{-3} for fragment analysis 1 and 5 000 MP# m^{-3} for fragment analysis 2 (Table 1).

We thus observe that extrapolated $MP^{\#}_{1-5000\mu m}$ are higher for films than for fragment 1, contrary to the originally reported concentrations. Importantly, even though the two fragment samples reported different MP size ranges, used different bin sizes to report the data, and reported different MP concentrations, the extrapolated $MP^{\#}_{1-5000\mu m}$ are identical. In summary, this fictional example illustrates how to compare proverbial apples and oranges, by converting them all into bananas.

Unfortunately, real world published MP PSD histograms do not behave like ideal PSDs, and it has been observed that power law slope depends directly on the chosen PSD bin size (Leusch et al., 2023). That is dramatic because an inaccurate slope directly leads to erroneous MP concentration extrapolation using Eq.(2), and it makes it impossible to compare slopes between datasets to understand MP size sorting in nature. If the raw data is available, it is possible to fit the power law to an unbinned, raw particle number PSD (Clauset et al., 2009) as applied by Kooi et al., (Kooi et al., 2021). However, since most published MP observations do not provide this raw data, we have developed an alternative method to overcome binning bias.

3. Binning bias

3.1. Datasets of reference

To further evaluate the effect of bin size on the BN-PSD slope, we obtained three raw MP number data sets from Lu et al. (2022), Perera et al. (2022) and Ueda et al. (2025), for stormwater, indoor air, and coastal marine surface water respectively. By raw data we mean that the authors reported the length and other dimensions of each individual MP particle observed. For Ueda et al. (2025), we selected the sub-dataset with the most particles (Fragment in Site D, sampled with a pump system). The Lu et al. (2022) and Perera et al. (2022) datasets were provided by Leusch et al. (2023). To obtain a reference slope and intercept for each dataset, we use the Clauset et al. (2009) maximum likelihood estimation (MLE) method that derives a best power law fit to raw particle size data. The MLE reference slopes are -1.75, -1.90 and -2.94 for Lu et al. (2022), Perera et al. (2022) and Ueda et al. (2025) respectively (see Figure 3.a and Table 2). The MLE intercept is calculated according to the formula provided by Clauset et al. (2009):

$$b = -(a + 1)\, L_{min}^{-(a+1)} \qquad \text{Eq.(3)}$$

where L_{min} is the lower size bound of the power law, i.e. the lower MP size limit for which the power law applies. An estimation of L_{min} is provided by the MLE algorithm (Clauset et al., 2009).

3.2. Bin-Normalized PSD method

Once the reference parameters are known for each of the 3 raw datasets, we apply to each of them the classic BN-PSD method. First, we generated binned PSD histograms with bin size incrementing from 1µm to 100 µm for each raw data set. We then plot the BN-PSD (x = log of the geometric mean size of each bin, y = log of the bin MP concentration divided by the bin width) and use standard linear regression to derive power law slope (*a*) and intercept (*b*). We use these parameters to extrapolate the observed MP# concentrations to the full 1-5000 µm MP size range. Table 2 summarises the parameters obtained as a function of bin size. Bin size does not affect every dataset in the same way. Figure 3.b clearly illustrates how the BN-PSD power law slope for all three datasets is systematically higher (less negative) than the MLE reference slopes. Consequently, the BN-PSD intercept and extrapolated $MP^{\#}_{1-5000\mu m}$ concentrations are also offset and therefore inaccurate. The Lu et al. (2022) et Ueda et al. (2025) slopes are relatively invariant with bin size, with an average RSD of 6% and 5% respectively. The BN-PSD method overestimated the Lu et al. (2022) et Ueda et al. (2025) MLE reference slopes by +0.28 (+16%) and +0.36 (+12%) respectively. In the case of Perera et al. (2022), the bin size used to aggregate data had a strong impact on the slope by the BN-PSD method: at bin size 1μm, the slope was -0.05, and this estimate decreases to -1.46 for 100 μm bins (Table 2, Figure 3.a). As bin size increases, the BN-PSD slope estimate gets closer to the MLE estimation. No regression was performed if the number of points was less than 3. The observed BN-PSD deviations from the theoretical MLE slope (Figure 3.a) propagate to the estimation of the intercept (Figure 3.b) and to the extrapolated concentration (Figure 3.c). Knowing that bin dependence and bias may arise in any binned dataset without warning signs, we explore in the next section an alternative method to derive accurate power law fits for binned data.

Table 2. *Average estimated parameters (slope, log intercept, log $MP_{1\text{-}5000\mu m}$ concentration) for three power law PSD fitting methods: bin-normalized particle size distribution (BN-PSD) fits the bin-normalized MP concentration against the geometric bin size in log log space (bin width varied from 1-100 μm); C-PSD fits the cumulative MP concentration against the lower bin bound (bin width varied from 1-100 μm) in log log space; MLE (maximum likelihood estimation) fits the raw unbinned data following Clauset et al. (2009) and gives an estimate of the parameters a and b independent of bin size. Values are averaged over all bin sizes tested, and uncertainties are 1 standard deviation.*

Parameter	Method	Lu et al. (2022)	Perera et al. (2022)	Ueda et al. (2025)
slope (a)	BN-PSD	-1.47±0.09	-0.81±0.36	-2.6±0.14
	C-PSD	-1.63±0.02	-1.64±0.02	-2.88±0.05
	MLE	*-1.75*	*-1.9*	*-2.94*
log intercept ($\log_{10}(b)$)	BN-PSD	4.1±0.22	1.12±0.82	5.83±0.28
	C-PSD	4.48±0.05	3.26±0.06	6.38±0.11
	MLE	*4.71*	*3.84*	*6.47*
$\log_{10}$ $MP^{\#}_{1-5000\mu m}$	BN-PSD	4.42±0.14	2.58±0.25	5.63±0.24
	C-PSD	4.68±0.04	3.45±0.04	6.11±0.1
	MLE	*4.84*	*3.88*	*6.18*

3.3. Cumulative PSD method

One alternative method to the BN-PSD is apparent in Virkar and Clauset (2014, Figure 1) who showed that power law fitting of a cumulative binned PSD (C-PSD) returns acceptably precise and accurate slope values, similar to the MLE. The C-PSD curve corresponds to the number (or concentration) of MP with length greater than L_1. In a C-PSD the number of MP counts (or number concentrations) is added up in each successively smaller bin, and the obtained concentration is plotted against the lower bound of each bin in log-log space. The resulting distribution is linear over most of the PSD, and far less sensitive to bin size. The C-PSD is defined as:

$$MP^{\#}_{>L_1} = \int_{L_1}^{+\infty} N(L)\, dL = \frac{-b}{a+1} {L_1}^{a+1} = b' {L_1}^{a'} \quad \text{Eq.(4)}$$

where $MP^{\#}_{>L_1}$ [MP# m^{-3}] is the cumulative number concentration of MP whose size is greater than L_1 [µm] (lower size of any bins). In log-log space, the C-PSD also follows a power law of slope $a' = a + 1$ (meaning the original BN-PSD slope is shifted by +1) and intercept of $b' = -b/(a+1)$. The parameters a and b are computed from the slope a' and log-intercept $log_{10}(b')$ obtained from the linear regression of the C-PSD: $a = a' - 1$ and $b = -b'.a'$. The MP number concentration within the size range L_1 to L_2 [µm] is given by:

$$MP^{\#}_{L_1 - L_2 \mu m} = MP^{\#}_{>L_1} - MP^{\#}_{>L_2} = b' {L_1}^{a'} - b' {L_2}^{a'} = \frac{b}{a+1} \left({L_2}^{a+1} - {L_1}^{a+1} \right) \text{Eq.(5)}$$

which is the same equation as Eq.(2) once a' and b' are converted to a and b. The only difference is the way the parameters a and b are estimated, from the C-PSD method for Eq.(5) rather than the BN-PSD method for Eq.(2).

In order to validate the C-PSD power law fitting method and compare its performance with the biased BN-PSD method, we used the same 3 reference raw MP datasets presented in section 3.1. Once again, we generated for all datasets binned PSD histograms with bin size incrementing from 1 µm to 100 µm, and used standard linear regression to derive power law slope *a'* and intercept *b'* of the C-PSD (*x*: log of lower bin bound, *y* = log of MP number concentration of MP > *x*) for each bin size. Figures 3.b-d and Table 2 summarize the validation results for slope, intercept and $MP^{\#}_{1-5000\mu m}$ concentrations. The C-PSD method gives a stable estimation of the slope, independent of bin size, with RSD <1.8%. The C-PSD slope is close to the MLE reference slope, deviating by only +0.12 (+7%), +0.26 (+14%) and +0.08 (+2%) for Lu et al. (2022), Perera et al. (2022) and Ueda et al. (2025) respectively. Similar acceptable accuracy is visible for the log intercept and extrapolated $MP^{\#}_{1-5000\mu m}$. It is interesting to note that the Perera et al. (2022) data are no longer sensitive to bin size. In summary, we find that the BN-PSD method alone is insufficient to correct bin-bias for all datasets, while the cumulative C-PSD fitting method returns more accurate slopes, intercepts and extrapolated MP concentrations. Next, we will outline how to convert MP number to mass concentration using the fitting parameters obtained by the C-PSD method.

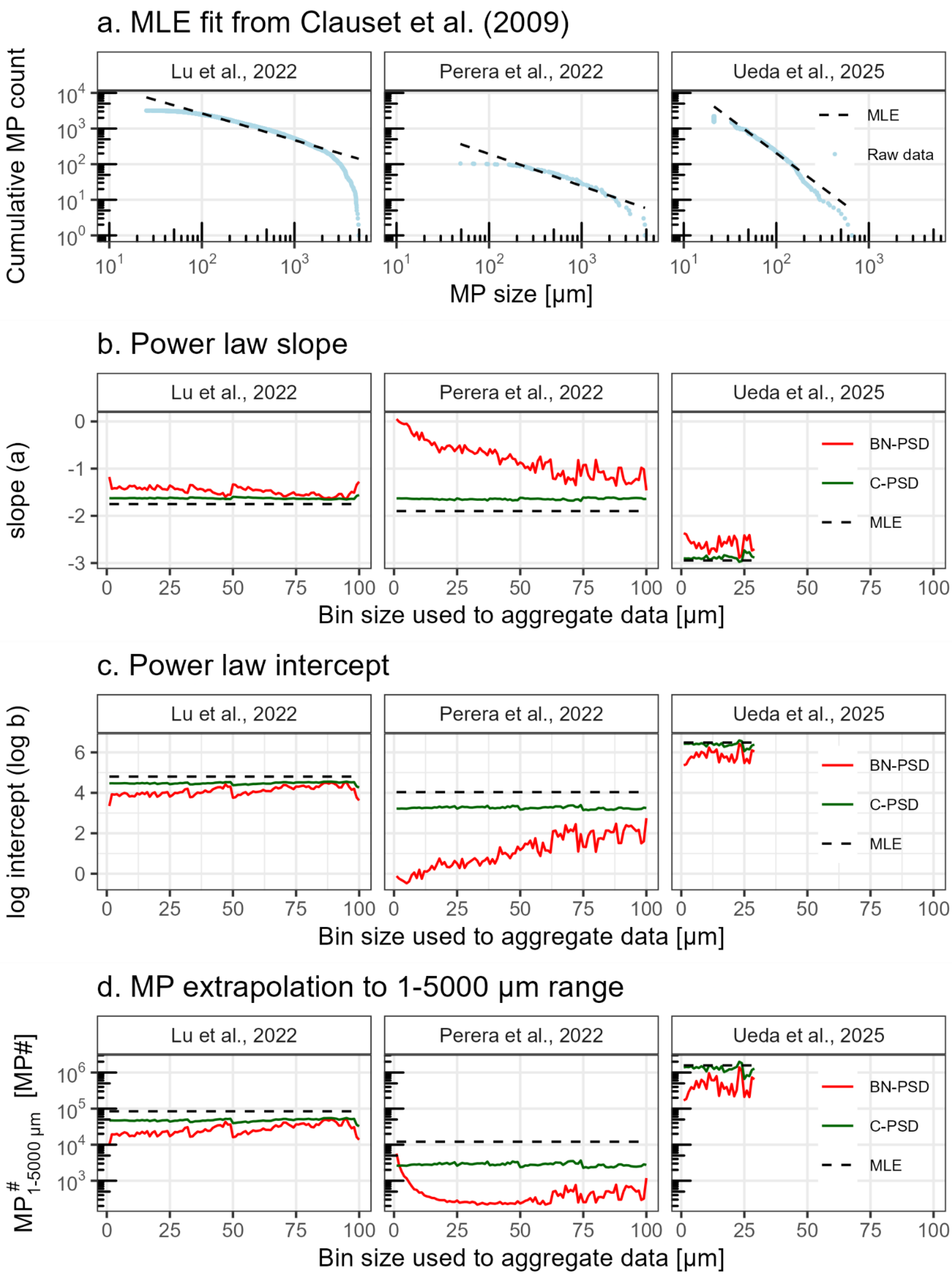

Figure 3. *Examples of binning bias for three raw MP datasets:* Lu et al. (2022) *and* Perera et al. (2022) *were discussed and provided by Leusch et al.* (Leusch et al., 2023). *For Ueda et al. (2025), we selected the subsample 'site D' with the most MP fragments obtained by a water pump. a) Visualisation of the maximum likelihood estimation (MLE) fit from Clauset et al. (2009) on the 3 raw datasets. The BN-PSD (in red) and C-PSD (in green) fitting methods*

presented in section 3.2 and 3.3 were tested as a function of bin size from 1 to 100 µm, and compared to the MLE (dotted lines) reference fitting method. The binning bias is illustrated in b) for the slope, in c) for the intercept and in d) for the extrapolated concentration $MP^{\#}_{1-5000\mu m}$.

4. MP number to mass conversion

MP mass concentrations, mass inventories and mass fluxes are needed for environmental plastic and MP dispersal budgets (OECD, 2022; Sonke et al., 2025a), for plastic additive exposure estimates (Trasande et al., 2024), or to understand plastic fragmentation and degradation (Chamas et al., 2020). To convert a MP number concentration to mass, the particle density and volume need to be known (or assumed) for the measured MP. While polymer density varies relatively little (0.9 to 1.4 × 10^{-6} µg/µm^3 (Kooi and Koelmans, 2019)), MP volume depends on MP length (L), width (W) –both typically measured– and height (H), which is generally not measured. The majority of studies unfortunately only report a PSD for L and omit W. Dedicated MP morphological studies typically observe that the median fragment W/L ratio is 0.68 +/- 0.03, indicating ellipsoid MP fragment shape (Barchiesi et al., 2023; Contreras et al., 2024; Hagelskjær et al., 2025; Kooi et al., 2021; Primpke et al., 2020a; Simon et al., 2018), and it has often been further assumed that $H/W = W/L$ (Kooi et al., 2021; Primpke et al., 2020a; Simon et al., 2018). Here we used a recent calibration of small fragment ellipsoid volume for the special case when only L is known, and which found H/W = 0.40 +/- 0.08 (Hagelskjær et al., 2025). We therefore convert number concentrations to mass concentrations using an average MP density of $\delta_{MP} = 1 \times 10^{-6}$ µg µm^{-3}, and an ellipsoid volume, V, approximation for fragments:

$$V_{frag}(L) = \frac{\pi}{6} \cdot L \cdot W \cdot H = \frac{\pi}{6} \cdot L \cdot 0.68\,L \cdot 0.4 \cdot 0.67\,L = 0.097\,L^3 \qquad \text{Eq.(6)}$$

Where L is the reported length (µm), and $W = 0.68 \times L$ and $H = 0.40 \times W$ (Hagelskjær et al., 2025).

For fibers, a cylinder volume approximation is computed using the reported diameter, *D* and a 40% void fraction (Simon18, Barchiesi23). If *D* is not reported, it is set to the typical value of 15 µm (Mintenig et al., 2020), except for rare fibers shorter than 45µm, where *D = L/3*, derived from the aspect ratio definition of fibers, *L/W>3,* used in several studies (Koelmans et al., 2022; Kooi et al., 2021; Ueda et al., 2025):

$$V_{fiber}(L) = \pi \cdot (D/2)^2 \cdot L \cdot 0.6 \qquad \text{Eq.(7)}$$

For films we assume a thickness (height, *H*) of 12.7 µm, corresponding to the typical grocery bag thickness:

$$V_{film}(L) = L \cdot W \cdot H \qquad \text{Eq.(8)}$$

The mass concentration (µg m^{-3}) of each shape category can then be estimated using the power law parameters *a* and *b* obtained by the C-PSD method described in 3.3:

Fragments:

$$MP^{mass}_{\tau_1-\tau_2\mu m} = \int_{L_1}^{L_2} \delta_{MP}\, V_{frag}(L)\, N(L)\, dL = \delta_{MP}\, 0.097\, \frac{b}{a+4}\left(L_2{}^{a+4} - L_1{}^{a+4}\right) \qquad \text{Eq.(9)}$$

Films:

$$MP^{mass}_{L_1-L_2\mu m} = \int_{L_1}^{L_2} \delta_{MP}\, V_{film}(L)\, N(L)\, dL = \delta_{MP}\, H\, \frac{b}{a+3}\left(L_2{}^{a+3} - L_1{}^{a+3}\right) \qquad \text{Eq.(10)}$$

Fibers:

$$MP^{mass}_{L_1-L_2\mu m} = \int_{L_1}^{L_2} \delta_{MP}\, V_{fiber}(L)\, N(L)\, dL = \delta_{MP}\, \pi\, 0.6\, \left(\frac{D}{2}\right)^2 \frac{b}{a+2}\left(L_2{}^{a+2} - L_1{}^{a+2}\right)$$

Eq.(11)

Where $MP^{mass}_{L_1-L_2\mu m}$ [µg m^{-3}] is the MP mass concentration within size range L_1 to L_2 [µm]. a and b are the slope and the intercept derived from a' and b' of the C-PSD power law model, respectively.

5. PSDs of environmental MP

In this section we examine the PSD properties of published MP studies, with a focus on different sampling and analysis methods, limits of detection and method bias and intercomparability. We use environmental MP data from the MPsizeBase compilation (Sonke et al., 2025b) that includes 62 binned MP PSDs for fragments and fibers from 40 studies in air (both indoor and outdoor suspended MP), surface ocean and subsurface ocean waters. Unlike the fictional example presented in section 1, environmental MP data rarely follow the power law perfectly over their whole reported PSD range. Authors generally do not formally define their limit of detection (LOD) and report all MP counts made in a single sample. This often generates lower than expected MP counts for the smaller PSD bins, and zero or single MP counts for upper size bins. Before fitting a MP PSD it is therefore important to consider what the analytical LOD on the lower MP size end (lower LOD), and on the upper MP size end (upper LOD) is.

5.1. Lower size LOD

When lower than expected small MP counts were initially documented and discussed by Cozar et al. (2014), it was interpreted as a true decrease in small MP abundance in ocean surface waters. Since then 'log-normal' PSD behavior has been observed in many MP datasets, and has been suggested to be related to sampling net mesh size (Lindeque et al., 2020; Tokai et al., 2021; Yu et al., 2025), to fragmentation energy barriers (Aoki and Furue, 2021; George et al., 2024) or to a low bias in MP counting, which is inherent in all measurement techniques, whether visual, software assisted, or microscope assisted (Kooi and Koelmans, 2019; Leusch et al., 2023; Primpke et al., 2020b). In Figure 4 we illustrate the log-normal PSDs of different environmental MP datasets obtained by microscopy (Cózar et al., 2014, 2015), by µFTIR (Eo et al., 2021; Isobe et al., 2015; Song et al., 2018; Tokai et al., 2021; Zhao et al., 2022), and by µRaman (Enders et al., 2015). The overlap of the observed MP size range between sampling and detection techniques allows us to visualise that the power law inflection point is dependent on the dataset and analytical technique used. This supports the assumption that the log-normal PSD phenomena is likely a low bias in the recovery and counting of small MP that exists for all sampling and detection techniques. Following Kooi and Koelmans (2019) and Kooi et al. (2021), we filtered the MPsizeBase data before fitting power laws to each study, by removing biased small MP counts. We first identify the bin of the BN-PSD with the maximum MP number concentration ($MP\#_{max}$, maximum on a BN-PSD curve). To conservatively exclude bins that underestimate MP count (which might be bins of size larger than the size of $MP\#_{max}$), the lower size LOD is defined as twice the geometric mean size of the bin that defines $MP\#_{max}$ (Figure 4.b). This condition was not applied when no visible low bias was observed.

It is important to emphasize that our attribution of log-normal MP PSDs behaviour to method bias is an important assumption of the power law C-PSD size alignment framework. PSDs for natural particles such as coarse mineral dust, seasalt aerosols or marine suspended particulate matter often display log-normal behavior below 10µm in size (Adebiyi et al., 2023; Deike et al., 2022; Stemmann et al., 2008). Aerosol size distribution in the 1 nm – 10 µm range is indeed understood as a superposition of multiple log-normal PSDs related to nucleation, accumulation and coarse modes (Deike et al., 2022). However, rarely do such PSDs show 10-fold drops in particle abundance as we illustrate for MP in Figure 4. As we will see later (Figure 5), only five MP PSDs reported in the MPSizeBase dataset have data <10 µm, and none of these show pronounced log-normal behavior. We therefore consider our assumption that the power applies over the full 1–5000 µm MP size range justified, but we remind readers that as higher quality data in the 1–20 µm range becomes available, we may need to revisit the possibility of log-normal MP PSD behavior.

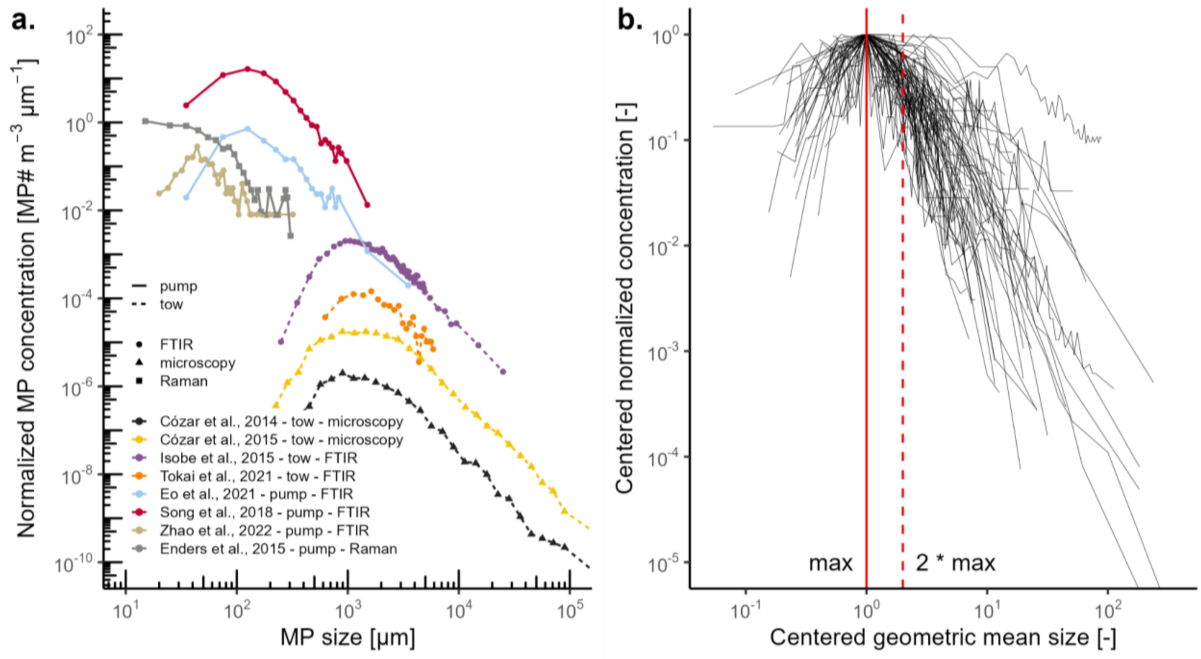

Figure 4. *a) Examples of BN-PSD inflection points for different sampling and analytical techniques (all surface ocean water, BN-PSD visualisation). b) Overview of all MPsizeBase PSDs, centered on the BN-PSD maximum MP# concentration (MP#$_{max}$) to illustrate residual non-linearity beyond MP#$_{max}$, and the assignment of a '2*MP#$_{max}$' lower size LOD.*

5.2. Upper size LOD

MP measurements generally detect fewer large particles, and at some point the number of large MP counts are no longer representative of the true large MP concentration. The question is at what size the upper size LOD is situated. The upper LOD of the BN-PSD and C-PSD is related to the sampling volume and the % of the sample (i.e. filter area) analyzed. In addition, for the C-PSD the right hand tail of the C-PSD slumps downward (becoming log-normal) as a function of the sum of all large particles that have not been counted. For fitting purposes, we assign the upper LOD to bins that define the downward slump breaking point in the log-linear nature of the C-PSD, which is often for bins containing a small fraction of the total MP counts. To illustrate the variability in reported MP size range, Supporting Information 1 shows for each of the 62 PSDs reported in MPsizeBase, the minimum and maximum MP size reported, and the lower and upper size LOD assigned. All 62 regressions are reported in Supporting Information 2 and visualize the lower and upper LOD assignment, alongside both BN-PSD and C-PSD regression lines. About half of the studies show log-normal behavior, and have a lower size LOD that is larger than the minimum MP size reported.

5.3. Environmental PSD slope variability

Figure 5 visualises MPsizeBase C-PSDs for each environmental compartment (atmosphere, surface and subsurface ocean) after lower and upper LOD assignment. To identify potential differences in power law slopes between MP shape and compartments, a two way Anova was performed (Figure 6). No statistical differences in power law slope (a) were identified between compartments (within fragment or fiber groups), for both number and mass PSD. This means that the current MPsizeBase dataset is not able to distinguish differences in PSD between atmosphere, surface ocean and deep ocean, indicating no detectable MP size sorting during sinking or emission and atmospheric deposition. We suspect that such differences in power law slope between compartments should exist, but would need more reliable data for it to be detected statistically. Conversely, we observe a strong statistical difference between MP fiber and fragment shapes ($p<0.0001$, Figure 6). For fibers, slope a varies from -1.30 to -2.59 with a mean value of -1.84 ± 0.38 (1 sd), and for fragments, slope a varies from -1.67 to -4.31 with a mean value of -2.76 ± 0.62 (1 sd). Compared to the theoretical slopes of the fractal fragmentation model for 1d, 2d and 3d plastic objects (section 1), this suggests that fragments behave close to ideal 3d fragmentation. Fibers, with slope of -1.84, however fragment more like

2d sheets than 1d fibers, which is surprising and may partly reflect the inclusion of fragment-like fibers in reported fiber datasets, with aspect ratio only slightly larger than 3.

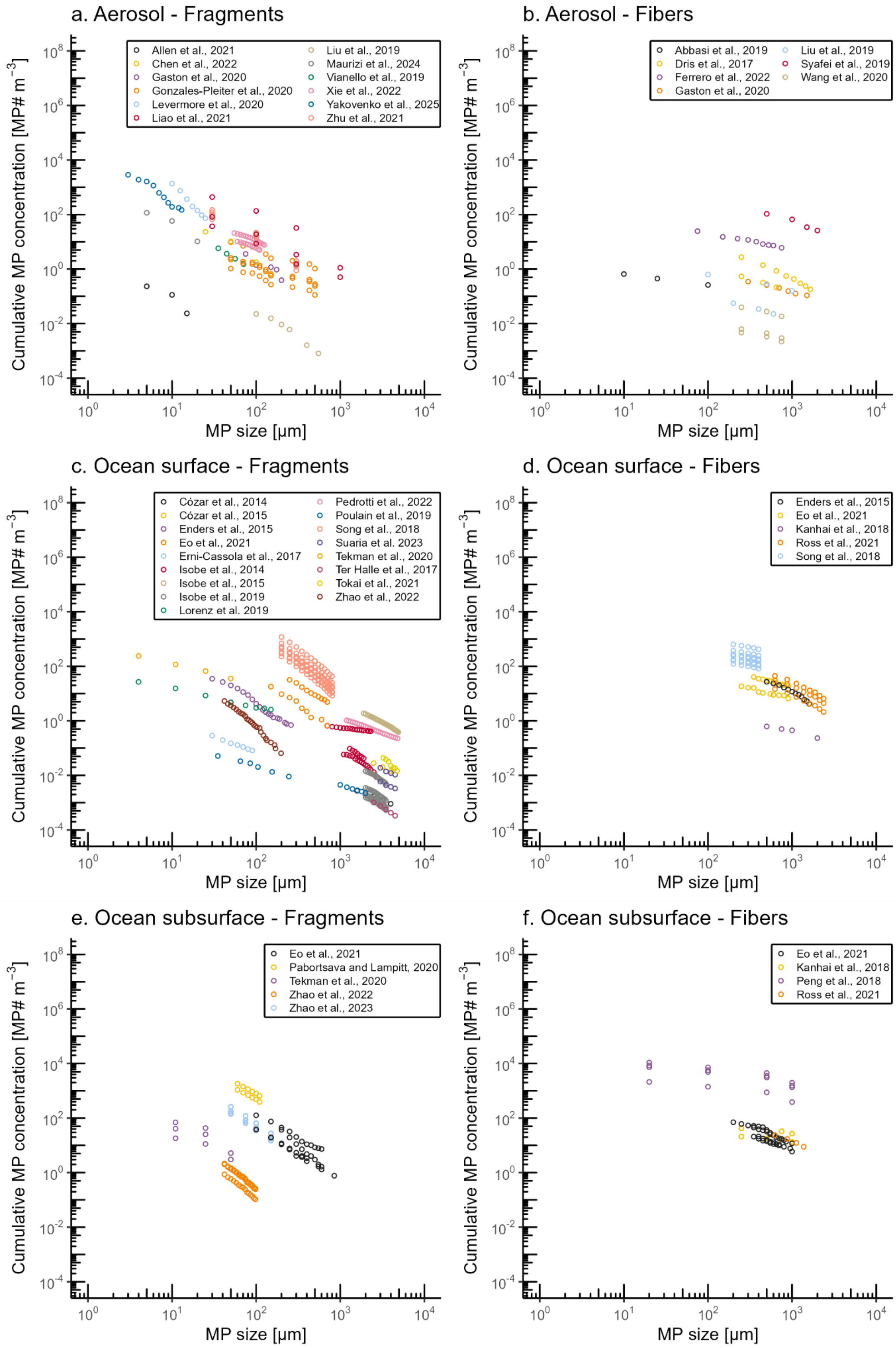

Figure 5 *: Published cumulative MP number particle size distribution (C-PSD) in atmosphere (a., b.), surface ocean (c., d.) and subsurface ocean (e., f.). The MP size (µm, lower bound of each reported bin) is plotted against the cumulative MP number concentration (MP# m^{-3}), which is the sum of reported MP concentration whose size is greater than x. C-PSD were separated into fragments (left panels, a., c., e.) and fibers (right panels, b., d., f.). Only data points within the assigned lower and upper LOD are shown (see 5.1 and 5.2). All individual C-PSDs are visualised in Additional file 2, including slope fitting and all points (including point removed for fitting).*

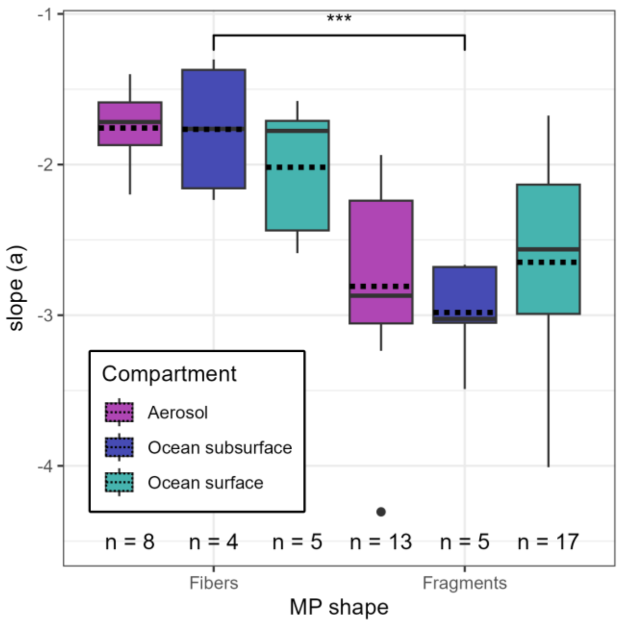

Figure 6 *: Two-factor Anova comparison of MP number concentration PSD power law slope between MP shape (fibers or fragments) and environmental compartments (atmosphere, ocean surface water or subsurface water). In MPsizeBase, the same PSD is often used several times with different concentrations (representing several sample locations or depths that have the same reported PSD), leading to duplicates in slopes. To avoid bias in the analysis, only one copy of each PSD was considered. In order to give each study equal weight, the median of the*

slope was taken for each study that reported more than one PSD for the same shape-compartment category. Full lines are medians, dotted lines are means. We observe a significant difference ($p = 1.7\ 10^{-6}$) in slope between the two MP shapes: -1.84±0.38 *(mean ± 1 sd) for fibers and* -2.76±0.62 *(mean ± 1 sd) for fragments. No statistical difference was observed between compartments, within each MP shape.*

Previous studies have examined PSD slope variability in different datasets. Initially, Kooi and Koelmans (2019) examined a mix of fiber and fragment PSDs, finding a mean slope *a* of -1.6 for number PSD. Later, Kooi et al. (2021) compiled raw MP data from five studies of marine surface water and sediment, freshwater and sediment, wastewater effluent and benthic freshwater biota. The five studies deployed different sampling methods, but comparable measurement techniques, including FTIR identification of MP using the same spectral reference database, >500 MP counts per sample, and >60% of filter surface analyzed. After filtering PSDs for lower LOD bias in small MP, the power law slopes, determined by the MLE method, for the predominantly (93%) fragment raw data were found to be very similar across all samples with a = -2.68 and -2.64 for length and width PSDs respectively. These fragment MLE slopes are, within their approximate uncertainty of ±0.6, not significantly different from our C-PSD derived fragment slopes of -2.76 ± 0.62 in atmospheric and ocean MP, indicating broad environmental similarity in MP PSDs across Earth surface environments. Kooi et al. (2021) observed minor statistically significant slope differences for MP length between surface water (a > -3) and biota (a < -3) in both freshwater and marine environments ($p<0.05$). Similarly, slopes were found to be >-3 for PE and PP, and <-3 for denser minority polymers in surface waters, suggesting preferential removal by settling of the larger, denser polymers. Kaandorp et al. (2021) highlighted potential differences in PSD shape for marine surface water MP at coastal and offshore locations studied by Isobe et al. (2014) and Isobe et al. (2015).

In summary, we observe different PSD for MP fragments and fibers that are to some extent coherent with the variable dimensionality of plastic fragmentation. It appears that the absence of statistically different PSD slopes between environmental compartments in the MPsizeBase data compilation is likely due to the great variety of sampling, MP counting, and MP reporting (i.e. binning) methods. The detection of minor, but significant slope differences in raw data PSDs measured with near-identical methods (Isobe et al., 2014, 2015; Kooi et al., 2021) suggests that standardization of measurement methods, and reporting raw data is key to tap into the full information on MP fragmentation and fractionation that is contained within PSDs.

Table 3. *Surface sea water MP fragment PSD comparison for pump and net tow sampling methods in coastal and offshore waters. Fiber PSDs are not shown.*

	N	N PSD	Size range	MP#/PSD	Volume sampled	PSD slope	PSD intercept	$MP^{\#}_{reported}$	$MP^{\#}_{l-5000\mu m}$	$MP^{mass}_{l-5000\mu m}$
			µm	MP#	m^3			MP# m^{-3}	MP# m^{-3}	µg m^{-3}
			reported	reported	reported			reported	estimated	estimated
			median	median (IQR)	median (IQR)	mean±1sd	median (IQR)	median (IQR)	median (IQR)	median (IQR)
Coastal										
Pump	2	10	35 - 1844	$4.5\ 10^4$ ($2.3\ 10^4$ - $6.8\ 10^4$)	0.15 (0.13 - 0.18)	-3.20±0.29	$2.2\ 10^8$ ($1.1\ 10^8$ - $3.3\ 10^8$)	716 (410-1021)	$9.1\ 10^7$ ($4.6\ 10^7$ - $1.4\ 10^8$)	6092 (3738-8446)
Net tow	5	7	250 - 4500	260 (184 - 3466)	543 (138 - 694)	-2.61±0.88	$3.8\ 10^5$ (48 - $1.1\ 10^6$)	0.41 (0.29-4.13)	$1.8\ 10^5$ ($7.1\ 10^1$ - $6.3\ 10^5$)	778 (163-3900)
Offshore										
Pump	3	3	15 - 310	239 (234 - 391)	0.39 (0.24 - 1.49)	-2.89±1.08	$1.1\ 10^5$ ($5.5\ 10^4$ - $5.3\ 10^5$)	54 (31-147)	$5.3\ 10^4$ ($2.7\ 10^4$ - $1.9\ 10^5$)	35 (18-2404)
Net tow	7	12	321 - 10 250	1100 (620 - 4070)	422 (231 - 510)	-2.42±0.37	2.7 103 ($5.6\ 10^2$ - $4.8\ 10^3$)	0.02 (0.01-0.05)	$1.9\ 10^3$ ($4.0\ 10^2$ - $3.2\ 10^3$)	10 (6-86)
All data										
Pump	5	13	20 - 320	494 (239 - 543)	0.20 (0.10 - 0.39)	-3.01±0.80	$9.4\ 10^5$ ($1.1\ 10^5$ - $1.7\ 10^6$)	105 (54-239)	$3.3\ 10^5$ ($5.3\ 10^4$ - $8.5\ 10^5$)	1383 (35-4772)
Net tow	12	19	286 - 5813	932 (253 - 4070)	452 (214 - 552)	-2.50±0.61	$3.2\ 10^3$ ($4.2\ 10^1$ - $1.0\ 10^5$)	0.17 (0.02-0.78)	$2.5\ 10^3$ ($5.8\ 10^1$ - $5.0\ 10^4$)	86 (6-966)

5.4. Sampling & analysis methods and PSD properties

In this section we examine how sampling and analysis methods affect the measured MP size range and MP concentration, and how C-PSD size-alignment can help reconcile perceived differences in MP counts.

5.4.1. Direct comparison of net tow vs pumped sampling

Surface water MP sampling is often done by towing a plankton, manta or neuston net with mesh size around 300 µm for 20–60 minutes. The large volume sampled, 200-600 m^3, is ideal to capture larger, less abundant MP in the 1–5 mm size range. These MP can be handpicked and manually counted and identified by spectroscopy, usually FTIR. More recent exploration of MP concentrations in the subsurface ocean relies mostly on discrete volume CTD casts (1–100 L), use of in situ pumps (100–800 L), filtration at a smaller mesh size of 1–20 µm, and automated FTIR or Raman particle finding or whole filter FTIR mapping techniques to identify small MP. In addition, shipboard pumps are now regularly used to sample surface water, again with lower volume (1–100 L) than plankton net tows. It has been recognized in the literature that net tows often report lower MP concentrations than pumped samples (Tamminga et al., 2019; Ueda et al., 2025) and we therefore ask the question whether that discrepancy disappears after MP size-alignment.

To investigate this we return to the unique Ueda et al. (2025) surface seawater study that directly intercompared neuston net tows (350 µm mesh) with discrete pumped (10 µm mesh) samples at four locations in the coastal, urbanized Tokyo Bay. Net tow MP were manually picked and analyzed by FTIR, while pumped MP were quantified by near-whole filter FTIR imaging. The reported mean MP concentrations for all sites were 430 (pump) and 0.59 MP# m^{-3} (net tow) for fibers, and 5989 (pump) and 0.75 MP# m^{-3} (net tow) for fragments. These reported MP# concentrations illustrate well the large apparent difference between pump and net tow methods, by factors of 730 (fibers) and 8000 (fragments). The authors indeed concluded that the pump system collected more small and fibrous MP, and that net tows were less effective in capturing MP <1000 µm. Figure 7 illustrates the pooled (all sites, A, B, C, D) cumulative MP concentrations (C-PSD, bin size 10 µm, filtered as described in section 5.1-2) for the 'net tow' and 'pump' groups, and for both fragments and fibers. For fragments (tow, n=3552; pump, n=292), the pump C-PSD in Figure 7b shows closer behavior to a power law than the net tow C-PSD which displays underestimation of small MP (thin black arrow). Nevertheless, it appears that large MP fragments (>1000 µm) from the net tow align well with the pumped MP size distribution, indicated by the dashed black lines (C-PSD slopes, Figure 7b) that are similar for both pump and tow C-PSDs. For fibers, both pump and net tow C-PSDs show curvature and underestimation of small MP (thin black arrows, Figure 7.a). The effect is stronger for tow than pumped samples, and no single power law fit can plausibly align both fiber distributions. Our visualizations confirm the authors' conclusion that at first sight, net tows appear to underestimate small MP concentrations.

Table 4 summarizes slope and intercepts for the merged Ueda et al. (2025) dataset, detailed by shape (fragments or fibers) and sampling method (net tow or pump), obtained with the MLE algorithm (Clauset et al., 2009) on raw data, and by the C-PSD method on binned (10 µm) data. Despite the reasonable visual alignment of tow and pump fragment C-PSDs, the MLE method generates significantly different slopes (Table 4). C-PSD fitting returns more coherent slopes and intercepts, which we use to extrapolate, using Eq. 2, MP number and mass concentrations over two default size ranges, $MP_{1\text{-}300\mu m}$ and $MP_{300\text{-}5000\mu m}$ (Table 4), in order to examine if extensive extrapolation is reasonable. The extrapolated $MP^{\#}_{300-5000\mu m}$ fiber concentrations are 79 MP# m^{-3} for pump and 18 MP# m^{-3} for net tow groups, indicating a 4-fold low bias for towed fiber sampling (grey arrow, Figure 7). Extrapolated $MP^{\#}_{1-300\mu m}$ fiber concentrations are 4.3 10^{4} MP# m^{-3} for pump and 3.7 10^{4} MP# m^{-3} for net tow groups. While this indicates good agreement, the underlying reason for this is that the slightly steeper net tow slope compensates for a pronounced low bias in measured small MP fibers. For fragments (Table 4), the extrapolated $MP^{\#}_{300-5000\mu m}$ fiber concentrations are 30 MP# m^{-3} for pump and 13 MP# m^{-3} for net tow groups, indicating agreement by a factor of 2 for larger MP fragments. For smaller MP fragments, extrapolated $MP^{\#}_{1-300\mu m}$ fiber concentrations are 2.7 10^6 MP# m^{-3} for pump and 2.3 10^5 MP# m^{-3} for net tow groups, indicating agreement to within a factor of 10. MP mass concentration extrapolation generally shows better agreement, because MP mass is dominated by the large MP size range that undergoes less extrapolation. Overall, the apparent 730-fold (fibers) and 8000-fold (fragments) difference in reported MP# concentrations between pump and net tow methods, is reduced to factors of 1-4 (fibers) and 2-10 (fragments) after size-alignment. This indicates that the apparent differences in reported MP# concentrations are mostly due to the different MP size ranges that were measured.

Table 4. *PSD fitting parameters for the Ueda et al. (2025) dataset. All 4 sites are merged together. MLE: maximum likelihood estimation on raw data (individual particle main axis); C-PSD: cumulative particle size distribution with bin size 10 µm and filtered as described in section 5.1-2.*

Sampling method	Fitting method	Slope (a)	Intercept (b)	$MP^{\#}_{reported}$	$MP^{\#}_{1-300\mu m}$	$MP^{\#}_{300-5000\mu m}$	$MP^{mass}_{1-300\mu m}$	$MP^{mass}_{300-500}$
				MP# m^{-3}	MP# m^{-3}	MP# m^{-3}	µg m^{-3}	µg m^{-3}
Fibers								
Pump	MLE	-2.14	6.0 10^4					
Net tow	MLE	-2.85	3.1 10^6					
Pump	C-PSD	-2.10	4.8 10^4	431	4.3 10^4	79	22	7.0
Net tow	C-PSD	-2.33	5.1 10^4	0.60	3.7 10^4	18	14	1.4
Fragments								
Pump	MLE	-3.23	2.1 10^7					
Net tow	MLE	-4.42	3.6 10^{11}					
Pump	C-PSD	-3.00	5.4 10^6	5989	2.7 10^6	30	159	2517
Net tow	C-PSD	-2.71	3.9 10^5	0.75	2.3 10^5	13	46	1673

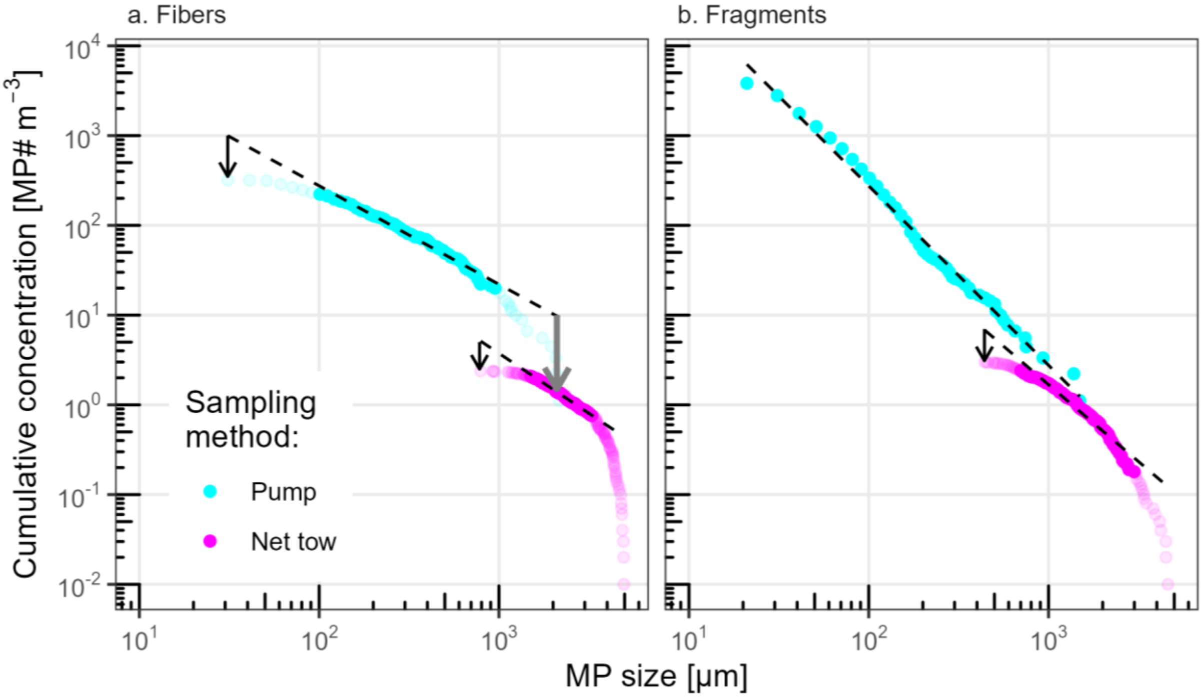

Figure 7. *PSD comparison between neuston net tow and pumped sampling methods in the Ueda et al. (2025) dataset for four Tokyo Bay surface waters. MP counts for all four sites (A-D) have been merged and converted to concentrations; the dashed lines are the C-PSD slope estimates for a bin size of 10 µm. Data was filtered as described in section 5.1-2 (transparent points below the lower LOD and above the upper LOD are excluded). a. For fibers both tow and pump PSDs are affected by low bias in small MP (thin black arrows), and the net tow PSD as a whole appears to have low bias in all MP fiber sizes (thick grey arrow) compared to the pump sampling method. b. For fragments a low bias in small MP sampled with the net tow is apparent from the non-linear shape of the pink PSD (thin black arrow).*

5.4.2 Indirect comparison of ocean surface water net tow vs pump data

We further examined PSD variability as a function of sampling techniques for surface ocean MP fragment (but not fiber) datasets in MPsizeBase: high volume net tows and low volume pumped samples (including surface CTD casts, in situ pumps, and shipboard pumps). Table 3 summarizes key statistics for the 'pump' and 'net tow' sample groups, with notably statistically different median sample volumes, 0.20 m^3 vs 452 m^3 respectively, and median reported MP size range, 20–320 µm vs 286–5813 µm respectively. Because the observed MP fragment size ranges are very different, the reported median MP number concentrations are also different, 105 vs 0.17 MP# m^{-3} for pump vs net tow groups. The key question is whether these reported MP concentrations align once we fit power laws and extrapolate to a common range, such as 1–5000 µm. After applying the lower and upper LOD data filters, the fitted C-PSD derived slopes, a, for the tow and pump groups are -2.50 ± 0.61 (n=12, 1sd) and -3.01 ± 0.80 (n=5, 1sd), and are not statistically different (t-test, p=0.16).

The reported tow and pump MP fragment concentrations of 0.17 and 105 MP m^{-3} are however confounded by coastal and off-shore sampling locations. Separating these two environments leads to reported median values of 0.41 (tow) and 716 MP# m^{-3} (pump) in coastal waters, and lower 0.02 (tow) and 54 MP# m^{-3} (pump) fragment concentrations in offshore waters. After C-PSD size alignment the extrapolated $MP^{\#}_{1-5000\mu m}$ fragment concentrations are 9.1 10^7 and 1.8 10^5 MP# m^{-3} in coastal waters, and 5.3 10^4 and 1.9 10^3 MP# m^{-3} in offshore waters for pump and net tow respectively. The very large apparent difference in reported marine MP fragment concentrations by pumped and towed sampling method (by factors of 716/0.41 = 1750, and by 54/0.02 = 2700) is reduced to factors of 500 and 30 following size alignment. Similar conclusions can be drawn for size-aligned median $MP^{mass}_{1-5000\mu m}$ fragment concentrations that extrapolate to 778 (tow) and 6092 ug m^{-3} (pump) in coastal waters, and 10 and 35 ug m^{-3} in offshore waters. Note, however, that these size-aligned pump and tow MP fragment concentrations are from different studies and geographical locations, and therefore do not represent the same waters as in the Ueda et al. (2025) study illustrated in section 5.4.1.

It is instructive to use the generic surface ocean fragment PSD slope of -2.62+-0.72 (1 sd, Table. 5) to estimate the MP fragment number and mass fractions that net tow and pump methods cover. One can compute using Eq.(5) that typical plankton/neuston/manta nets (mesh size ~300 µm) used to sample surface ocean MP recover less than 1% of the total number of MP fragments in the size range 1–5000 µm. On the other hand, the same 300 µm mesh net catches 85% to 100% of the total floating MP fragment mass in the size range 1–5000 µm. Due to the exponential nature of the power law, about 98% of all surface ocean MP fragments are smaller than 10 µm. In the MPsizeBase database, only two studies (11%) observed MP fragments smaller than 10 µm at the ocean surface (Lorenz et al., 2019; Tekman et al., 2020).

5.4.3 Indoor airborne MP by FTIR and Raman

The importance of size alignment when intercomparing different MP measurement techniques can also be illustrated for FTIR and Raman quantification of indoor airborne MP fragments. In MPsizeBase three FTIR and three Raman studies detected 361 and 1510 MP# m^{-3} across MP size ranges of 18–400 µm and 3–75 µm respectively. The factor 9 concentration difference is mostly related to the capacity of Raman microscopy to detect more abundant smaller MP (Yakovenko et al., 2025). PSD slopes, *a*, and intercepts, *b*, were similar for FTIR and Raman: -2.71±0.50 vs -2.74±0.50 and 2.7 10^4 vs 1.8 10^4 respectively. The apparent 9-fold difference in reported MP fragment concentrations by FTIR and Raman becomes negligible after size alignment to the common 1–5000 µm range ($MP^{\#}_{1-5000\mu m}$): 16575 vs 16581 MP# m^{-3}.

6. Extrapolating and intercomparing environmental MP concentrations

6.1 Extrapolation of MPsizeBase datasets with C-PSD fitting

All 62 fitted MPsizeBase PSDs were extrapolated beyond the original PSD observations, and $MP_{1\text{-}5000\mu m}$ number and mass concentrations estimated using Eq.(5) for number and Eq.(9-11) for mass. The extrapolated concentrations are summarized in Table 5. Figure 8 compares reported measured MP concentrations over restricted size ranges to extrapolated $MP^{\#}_{1-5000\mu m}$ concentrations for the formal 1–5000 µm MP size range. It can be observed that extrapolated $MP^{\#}_{1-5000\mu m}$ number concentrations are always higher than measured MP number concentrations by a median factor of 600x (IQR: 30–8000). For fragments, the extrapolated $MP^{\#}_{1-5000\mu m}$ concentration is higher by a median factor of around 70x (IQR: 20–1000) for aerosols, 2000x (IQR: 1300–2300) for the subsurface ocean and 11000x (IQR: 700–150000) for the surface ocean MP. For fibers, these factors are 50x (IQR: 20–120) for aerosols, 300x (IQR: 60–700) for subsurface ocean and 80x (IQR: 40–6000) for surface ocean MP fibers. The reason for this is that only a sub range of the full MP size spectrum (1 and 5000 µm) is observed, which contains a small fraction of the total MP number. In particular, abundant small MP are not observed by FTIR studies. Fibers have lower extrapolation factors as their slopes tend to be less steep than fragments (Figure 6). For MP mass concentrations, the extrapolated $MP^{mass}_{1-5000\mu m}$ are higher by a median factor of 3x (IQR: 1.5–12) than the estimated measured concentrations. For many net tow datasets, mass extrapolation leads to only a limited increase in MP mass concentration, because large MP size ranges from 300 to 5000 µm, which dominate mass, are already covered by the original observed data.

6.2. Extrapolating MP concentration data without C-PSD fitting

Depending on research questions and objectives there may be needs for extrapolation of MP# concentration without going through the complexity of C-PSD fitting individual datasets. This can be done by using generic, mean power law slopes, for different environmental compartments. Koelmans and Kooi (2020) developed equations, similar to Eq.(2) above, for a MP concentration correction factor, CF, that allows extrapolation of a reported MP#

concentration to any size range, based on the minimum and maximum MP size observed, and a literature-based estimate of the generic PSD slope *a*:

$$CF = (D_{max})^{1+a} - (D_{min})^{1+a} / (L_{max})^{1+a} - (L_{min})^{1+a} \qquad \text{Eq.(12)}$$

, where L_{min} *and* L_{max} are the reported minimum and maximum MP length (µm), and D_{min} *and* D_{max} refer to the default size range of choice (e.g. 1–5000 µm). Corrected, extrapolated $MP^{\#}_{Dmin-Dmax}$ concentrations are:

$$MP^{\#}_{Dmin-Dmax} = MP^{\#}_{Lmin-Lmax} * CF \qquad \text{Eq.(13)}$$

In Table 5 we summarize our best estimates for generic PSD slopes, *a,* for fibers and fragments in the atmosphere, surface ocean and subsurface ocean. We note, however, two elements of uncertainty in this CF approach: i) About half of all reported PSDs show low bias for small MP (Figure 4b); consequently the reported measured MP# concentrations are also biased low, which is subsequently propagated to the extrapolated MP number concentration via Eq.(12). ii) Measured L_{min} and L_{max} are rather uncertain size estimates of observed MP populations, because they represent extreme values. An approximate L_{max} introduces little additional uncertainty during MP# extrapolation because extrapolated large MP counts are rare. On the lower L_{min} end this is not the case, and we recommend replacing L_{min} by the bias-corrected $L_{lowerLOD}$ bound. This, however, demands a corresponding correction of the biased reported MP# concentration that can be calculated in MPsizeBase. Further use of generic PSD slopes to extrapolate MP mass concentrations is best done by first calculating intercept *b* for each datapoint using Eq.(2), and then apply both generic slope *a* and datapoint-specific *b* using Eq's (9-11).

Database users that wish to extrapolate and size-align large amounts of MP number or mass concentration observations have the choice of using i) study-specific PSD and slope estimates, or ii) generic mean PSD slopes for different environments (Table 5). Further work would be needed to evaluate which approach is more appropriate. Study-specific PSD slope variability represents all method-related uncertainty, which is transferred and amplified to extrapolated MP concentrations. Use of a generic PSD slope, that represents the mean PSD properties of all methods, may result in lower uncertainty of extrapolated MP concentrations. Development and measurement of certified reference materials, and comparison of both approaches against MP dispersion models can help clarify the choice of extrapolation method.

7. Conclusions

We have extended the existing power law PSD framework for microplastic to extrapolate and size-align binned MP size and number concentration data in the literature. The advantage of the framework is that it facilitates data inter-comparison of observations and of observation-model output by comparing MP number or mass concentrations over the same MP size range. The main disadvantage is that the assumption of MP PSDs to follow a power law precludes detailed investigations of particle size fractionation processes, as current data quality does not result in significant PSD slope variability in atmospheric, surface and subsurface ocean.

We recommend future environmental MP studies to always provide raw MP particle size data (length, width, area, perimeter, circularity) in numerical format as part of the supporting information or in an online data repository (cf. submission to MPsizeBase). Raw data, unbinned, provides additional estimates of PSD slope and intercept. Binning MP size data is mostly useful for visualization of PSDs histograms in papers.

More studies should intercompare sampling and analytical methods on the same samples, so that we can better understand the strengths and limitations of methods. In the marine environment we show that net tow and discrete, pumped sampling methods provide potentially complementary concentration results for MP fragments. Both methods are important, because large volume net tows recover less abundant large MP that carry most MP mass, while low volume pumped samples allow analysis of small MP that are relevant for biota uptake, health impacts, sea spray emission to the atmosphere and carbon cycling perturbations. Nevertheless, more efforts should be made to observe small MP in the 1–20 μm range by Raman microscopy in all environments. Net tow sampling of MP fibers appears to substantially underestimate fiber concentrations, even for the larger size range. Dedicated methodological developments should address both sampling and detection of fibers in environmental media.

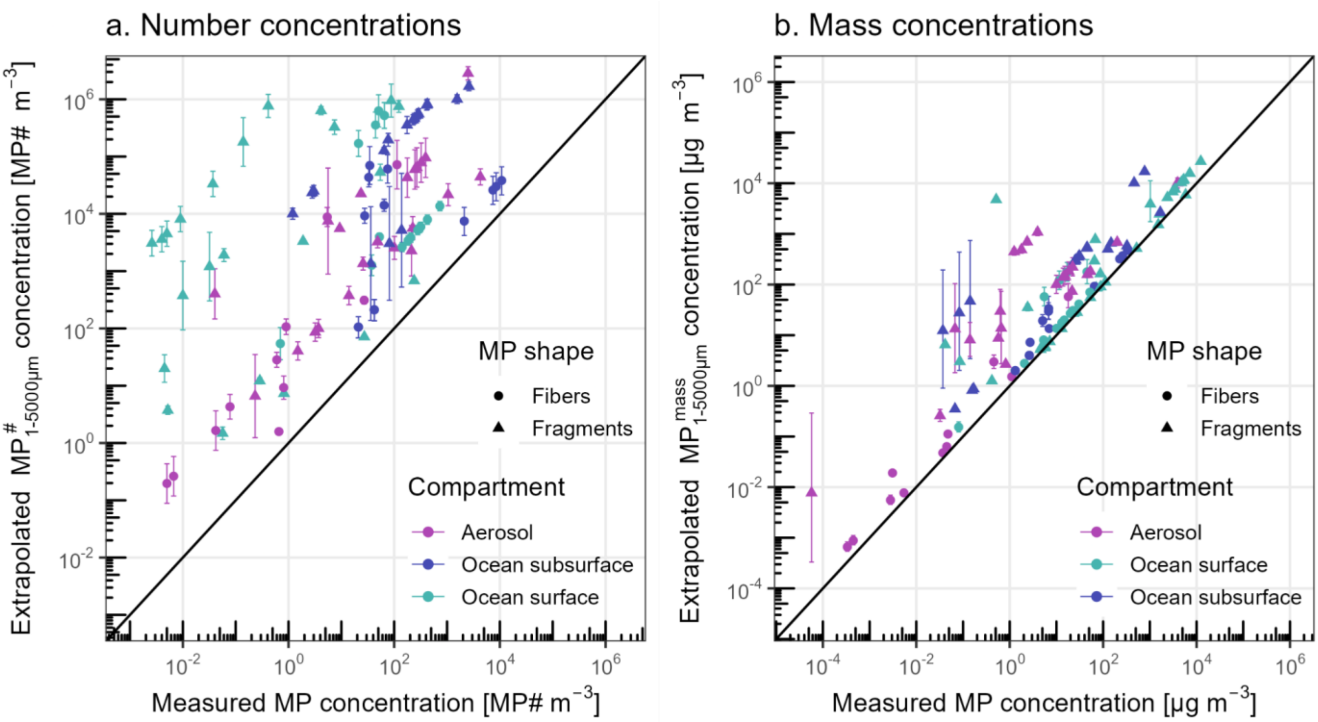

Figure 8. *Comparison of measured MP concentrations to extrapolated $MP_{1\text{-}5000\mu m}$ concentrations in a) number (MP# m^{-3}) and b) mass (μg m^{-3}). Error bars are estimated extrapolation uncertainties (1 standard deviation). Because only a fraction of the MP size spectrum between 1 and 5000 μm is measured, the extrapolated MP number and mass concentrations are generally higher than the corresponding reported concentrations. Data sampling dates cover the period 2011–2023.*

Table 5. *Summary of C-PSD size-aligned literature data by environmental compartment and MP shape. Values are mean±1 SD for normally distributed data, and median (IQR) for log-normally distributed data. N: number of MP data points; N*: number MP PSDs.*

	N (N*)	slope (a)	intercept (b)	$MP^{\#}_{reported}$	$MP^{\#}_{1-5000\mu m}$	$MP^{mass}_{1-5000\mu m}$
				MP# m^{-3}	MP# m^{-3}	µg m^{-3}
Ocean surface						
Fibers	16 (7)	-2.04±0.42	3169 (2047-5.64 10^5)	51 (40-53)	5443 (2618-3.92 10^5)	28 (5-133)
Fragments	32 (21)	-2.62±0.72	5862 (48-1.06 10^6)	0.4 (0.1-27)	3328 (71-6.34 10^5)	113 (6-1528)
Ocean subsurface						
Fibers	14 (7)	-1.92±0.41	1.39 10^4 (1.06 10^4-6.49 10^5)	42 (34-70)	2.81 10^4 (1.17 10^4-5.20 10^4)	19 (10-32)
Fragments	18 (8)	-3.00±0.42	4.86 10^5 (1.82 10^5-1.39 10^6)	116 (74-202)	2.76 10^5 (1.01 10^5-7.43 10^5)	542 (277-1161)
Aerosol						
Fibers	11 (9)	-1.81±0.31	22 (4-184)	0.8 (0.6-5)	28 (4-310)	0.06 (0.02-1.5)
Fragments	22 (14)	-2.76±0.61	7923 (2189-4.09 10^4)	37 (7-253)	4827 (1585-2.23 10^4)	22 (8-376)

Abbreviations

MP: microplastic; NP: nanoplastic; PSD: particles size distribution; BN-PSD: bin-normalized particle size distribution; C-PSD: cumulative particle size distribution; MLE: minimum likelihood estimation; LOD: limit of detection; FTIR: Fourier transform infra-red

Supplementary Information

The online version contains supplementary material available at [...]

Additional file 1: .xlsx file summarizing all the MPsizeBase dataset used in this study.

Additional file 2: .zip file with visualization of all PSD fighting with both BN-PSD and C-PSD method.

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Authors' contributions

JES and TS designed the study. JES, JLT and HA acquired funding. JES, TS, IH, DV, CR and ND developed the C-PSD size alignment framework. All authors contributed to data interpretation. JES, TS and IH wrote the draft manuscript, which was improved with the help of all authors.

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Availability of data and materials

The authors declare that the data supporting the findings of this study are available within the paper and its supplementary information files, as well as at https://zenodo.org/records/17380284.

Declarations

Ethics approval and consent to participate

NA.

Consent for publication

The authors provide consent for publication.

Competing interests

The authors declare no competing financial or other interests.

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